M|P|W

Physics A (H156, H556)

Ch.1-7_Revision TA_2026
The Physics Department

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:
Additional resources may be supplied with this paper.

Other materials required:
- Pencil
- Ruler (cm/mm)

Duration: 45 mins

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets **[ ]** at the end of each question or part question.
- The total number of marks for this paper is **40**.
- The total number of marks may take into account some 'either/or' question choices.

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1 The table below shows the measurements recorded by a student for a solid metal sphere. The absolute uncertainties in the mass of the sphere and in its radius are also shown.

mass	100 ± 6 g
radius	1.60 ± 0.08 cm

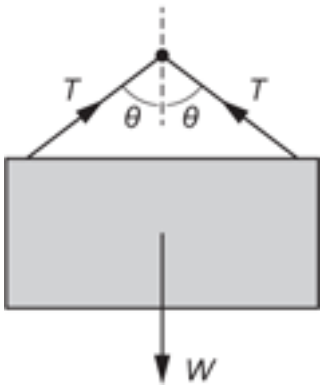
What is the percentage uncertainty in the density of the sphere?

A. 1%
B. 11%
C. 16%
D. 21%

Your answer

[1]

2 A cable is used to hang a picture from a nail. The diagram shows all the forces acting on the picture. T is the tension in the cable and W is the weight of the picture.



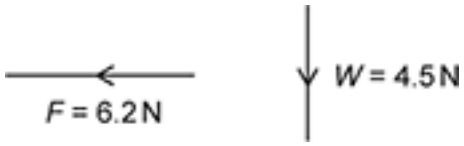
Which is the correct expression for W?

A $W = T \cos \theta$
B $W = 2T \cos \theta$
C $W = T \sin \theta$
D $W = 2T \sin \theta$

Your answer

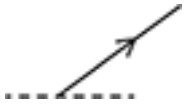
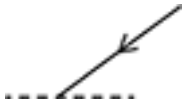
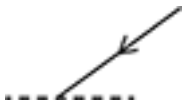
[1]

3 An object is in equilibrium under the action of three coplanar forces F, R and W. The diagram below shows the forces F and W.



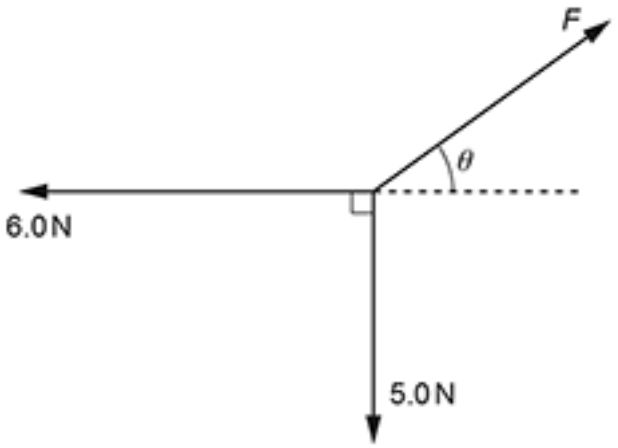
The angle between F and W is 90°.

Which row shows the correct magnitude of R and the approximate direction of R?

	Magnitude of R / N	Direction of R
A	7.7	
B	7.7	
C	10.7	
D	10.7	

Your answer ☐ **[1]**

4 A free-body force diagram is shown below. The system is in equilibrium.



Calculate the magnitude of the force F and the angle θ.

	F / N	θ / °
A	7.8	40
B	7.8	50
C	11	40
D	11	50

Your answer [1]

5 A ball, initially at rest, is struck by a hockey stick. It leaves the hockey stick at speed v.

Which quantity, together with the mass of the ball, can be used to determine v?

A The time of the impact.

B The weight of the hockey stick.

C The impulse of the force.

D The final momentum of the hockey stick.

Your answer [1]

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6 An object experiences two forces, 3.0 N and 4.0 N, in the same plane. The directions of the forces are not known.

What is the magnitude of the resultant force F acting on the object?

A $F = 5.0$ N

B $F = 7.0$ N

C $1.0 \text{ N} \leq F \leq 7.0 \text{ N}$

D $4.0 \text{ N} \leq F \leq 7.0 \text{ N}$

Your answer [1]

7 Two identical particles are travelling at the same speed v in opposite directions.

They collide head on. The collision is elastic.

What happens to the two particles after the collision?

A One particle stops and the other rebounds at speed $2v$.

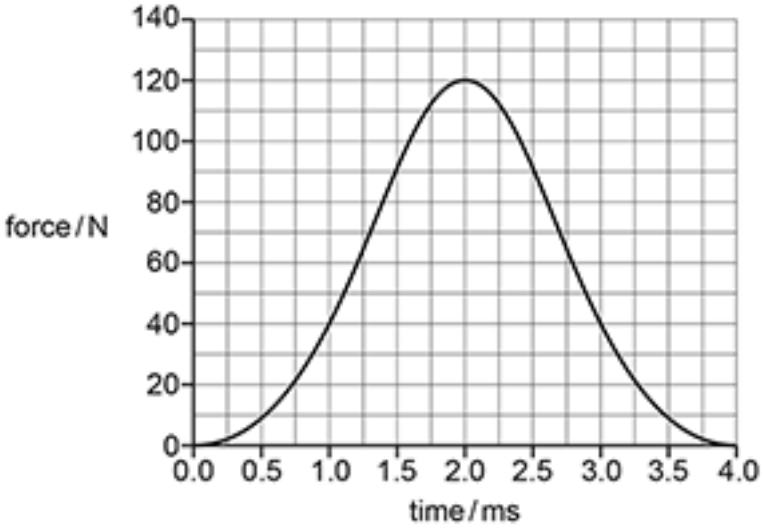
B They move off together at speed v.

C They rebound in opposite directions, each at speed $\frac{v}{2}$

D They rebound in opposite directions, each at speed v.

Your answer [1]

8 A ball is hit by a racket. The graph shows how the force on the ball changes whilst the racket is in contact with the ball.

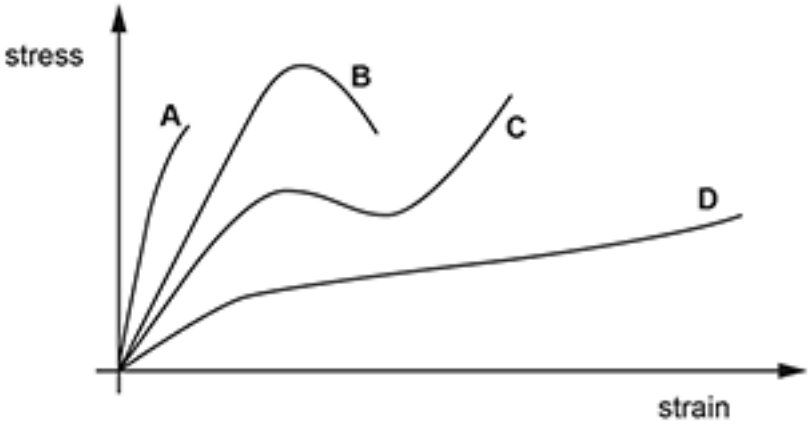


Which is the best estimate of the total impulse on the ball?

A 0.20 N s
B 0.24 N s
C 0.48 N s
D 0.50 N s

Your answer ☐ **[1]**

9 The stress-strain graphs for four materials **A**, **B**, **C** and **D** are shown.

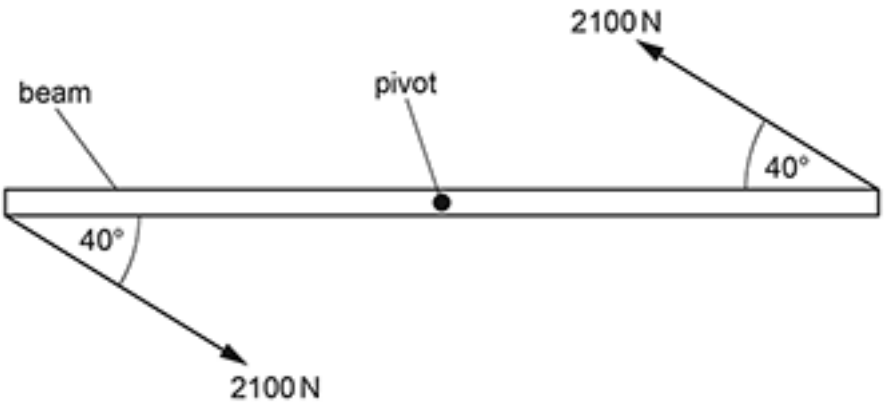


Which material has the greatest ultimate tensile strength?

Your answer ☐ **[1]**

10 A beam of length 4.0 m is pivoted at its centre.

Two equal and opposite forces of magnitude 2100 N are applied to opposite ends of the beam, as shown.



What is the torque due to the two forces shown?

A 2700 N m
B 3200 N m
C 5400 N m
D 6400 N m

Your answer [1]

11 A spring has a force constant of 4900 N m^{-1}.

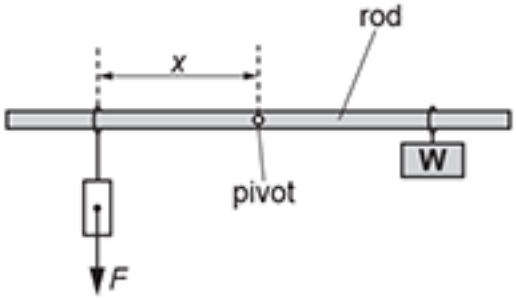
A force is applied to the spring, causing it to compress by 0.50 m.

What is the change in the elastic potential energy stored in the spring?

A decreases by 610 J
B decreases by 1200 J
C increases by 610 J
D increases by 1200 J

Your answer [1]

12 A student balances a uniform metal rod horizontally.



The rod is pivoted at its middle. The position of weight **W** is kept constant.
The distance of the weight *F* from the pivot is *x*. The student changes *F* and then adjusts *x* so that the rod remains balanced.
Which statement is correct?

A A graph of *F* against *x* will be a straight line through the origin.
B The upward force at the pivot is equal to *F*.
C The weight of **W** is equal to *Fx*.
D *x* is inversely proportional to *F*.

Your answer ☐ **[1]**

13 An electric motor is used to lift a weight of 4.0 N through a vertical height of 0.90 m in 1.8 s. The efficiency of the motor is 20%.
What is the electrical power supplied to the motor?

A 0.40 W
B 2.0 W
C 3.6 W
D 10 W

Your answer ☐ **[1]**

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14 In an experiment, a trapdoor and electromagnet are used to determine the acceleration of free fall of a ball.

The distance the ball falls is h and the time taken for the ball to fall is t.

The experiment is repeated for different values of h.

The table shows the results. Values of $\sqrt{h}$ have been included.

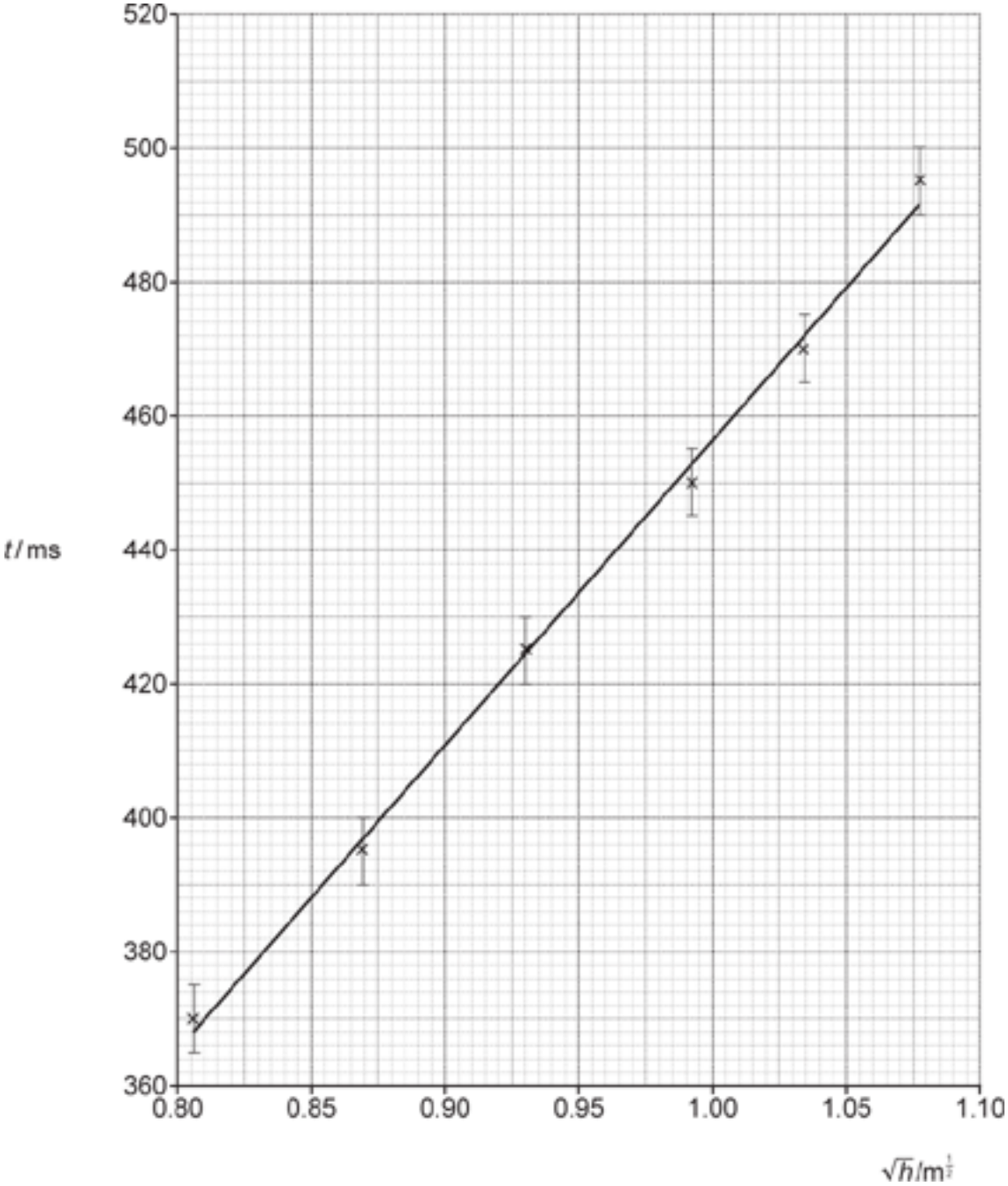
h/m	$\sqrt{h}$ / $m^{\frac{1}{2}}$	t/ms
0.650	0.806	370 ± 5
0.755	0.869	395 ± 5
0.865	0.930	425 ± 5
0.985	0.992	450 ± 5
1.070	1.034	470 ± 5
1.160	1.077	495 ± 5

It is suggested that the relationship between t and h is

$$t = \sqrt{\frac{2}{g}}\sqrt{h} + k$$

where g is the acceleration of free fall and k is a constant.

A graph of t / ms on the y-axis against $\sqrt{h}$ / $m^{\frac{1}{2}}$ on the x-axis is plotted.



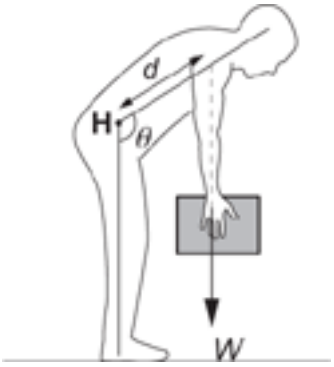
- Describe how to measure *h* and *t*.
- Use the graph to determine *g*, including the percentage uncertainty.

[6]

15(a) A lab technician is moving boxes.

The technician lifts a box from the floor without bending their knees.

The diagram shows the force *W* due to the weight of the box.



The box has a mass of 5 kg.

The distance *d* is 0.6 m and can be assumed to remain constant.

Calculate the moment about the point **H**, due to the weight of the box, when $\theta = 90^{\circ}$.

State the unit.

moment = ... unit.....[2]

(b) The diagrams show how the technician can pick up the box while bending their knees.

This keeps their spine more vertical.

Explain why bending the knees is less likely to cause damage to the spine.

[3]

i. Show that an expression for the Young modulus E of a wire is

$$E = \frac{kL}{A}$$

L = length of wire
A = cross-sectional area of wire
k = force constant of wire

[2]

ii. The student records the following results for a copper wire.

L = 2.0 m ± 0.05 %
A = 2.9×10^{-8} m^2 ± 2 %
k = 1670 N m^{-1} ± 1.25 %

Calculate the value of the Young modulus of the wire and its percentage uncertainty.

Young modulus = ...N m^{-2} ± .....% **[3]**

iii. The student researches the Young modulus of copper.

They find a value of 1.17×10^{11} N m^{-2}.

Determine whether this value is consistent with your answer to **(c)(ii)**.

[2]

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17 Two identical springs each have a force constant of 36 N m^{-1}. In an experiment, the two springs are suspended from a fixed support as shown in **Fig. 4.1**.

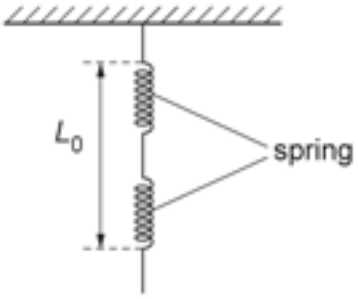

Fig. 4.1

The initial length of the spring arrangement is L_0.

An object of mass M is added to the spring arrangement as shown in **Fig. 4.2**.

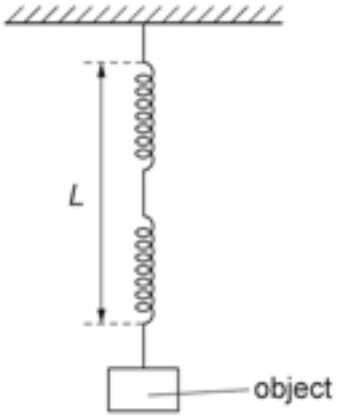

Fig. 4.2

The new length of the spring arrangement is L.

A student measures L_0 and L and records the results in a table.

Quantity	Measurement / mm
L_0	(22.2 ± 0.1)
L	(54.9 ± 0.1)

i. State the name of the instrument the student used to measure L_0 and L.

[1]

ii. Determine the extension x of the spring arrangement. Include the absolute uncertainty in your answer.

x = ...± ..... mm **[1]**

iii. Calculate the mass M of the object. Write your answer to **2** significant figures.

M = ...kg **[2]**

iv. Calculate the total energy W stored by the springs when the object is suspended.

W = ...J **[2]**

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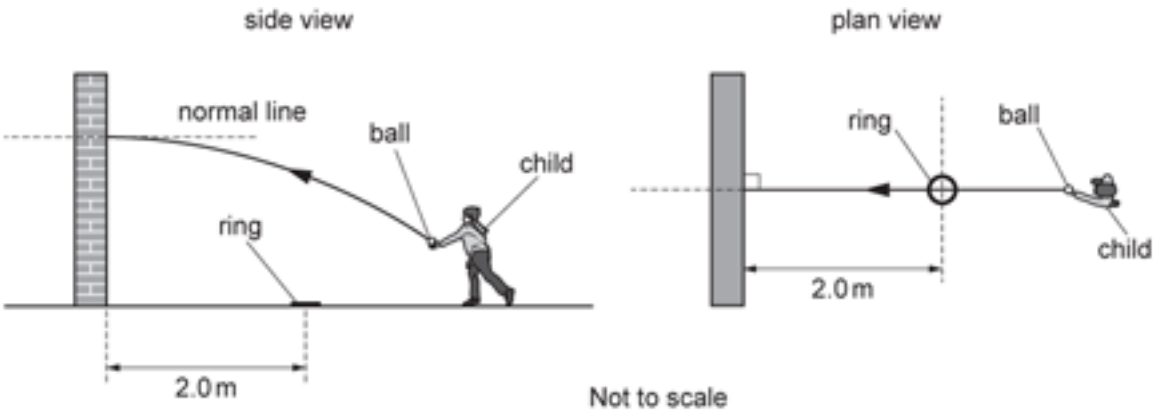
18 In a game, a child throws a ball at a flat, vertical wall. The ball rebounds from the wall. The child wins the game if the ball lands within a circular ring placed on the ground.



The ring has a radius of 15 cm. The centre of the ring is 2.0 m from the wall. The child throws a ball with a mass of 0.058 kg towards the wall.

The ball rebounds normally with a horizontal velocity of 3.6 m s^{-1} and lands on the ground a time of 0.58 s after hitting the wall.

Show that the child wins the game. Air resistance can be assumed to be negligible.

[3]

END OF QUESTION PAPER

Question			Answer/Indicative content	Marks	Guidance
1			D	1	
			Total	1	
2			B	1	<u>Examiner's Comments</u> This question was generally answered well as most candidates correctly resolved *T*, tension to determine an expression for *W*.
			Total	1	
3			A	1	
			Total	1	
4			A	1	<u>Examiner's Comments</u> A few candidates incorrectly selected C – adding the two forces 5.0 N and 6.0 N to equal 11 N.
			Total	1	
5			C	1	
			Total	1	
6			C	1	
			Total	1	

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Question			Answer/Indicative content	Marks	Guidance
7			D	1	<u>Examiner's Comments</u> The question was answered well. Candidates usually correctly identified that the particles would rebound in opposite directions but the common error was to suggest the speed of the particles would be halved. If this was the case, then the kinetic energy would not be conserved. Assessment for learning Candidates should understand in all collisions: • the total momentum is conserved, and • total energy is conserved. Candidates should understand that in an **elastic collision** • the total momentum and • the total energy is conserved and • the total kinetic energy is conserved.
			Total	1	

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Question			Answer/Indicative content	Marks	Guidance
8			A	1	<u>Examiner's Comments</u> Candidates found this question more challenging. Some candidates realised that the total impulse was given from the area under the graph but then did not calculate the area correctly. Some clearly drew a triangle and gained an answer of 0.24 N s without allowing for the area inside the triangle. Other candidates multiplied. Assessment for learning Candidates should understand how to calculate the area under a graph: • work out the physical significance of one small square • calculate a value for the one small square considering the units • count the number of whole small squares • estimate the total number of whole squares by allowing for the part squares • multiplying the total number of squares by the physical significance of one small square.
			Total	1	
9			B	1	<u>Examiner's Comments</u> The majority of the candidates correctly identified B. The common distractors were C and D.
			Total	1	

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Question			Answer/Indicative content	Marks	Guidance
10			C	1	<u>Examiner's Comments</u> The majority of the candidates who did not gain the correct answer, selected answer A. These candidates correctly calculated the force but found the moment of the pivot for one of the forces rather than the torque of the two forces shown.
			Total	1	
11			C	1	<u>Examiner's Comments</u> Overall, candidates performed well on this question with many correctly calculating the change in elastic potential energy as 610 J. The most common distractor was answer A, with candidates relating compression to a decrease in the change of elastic potential energy rather than an increase in the store of elastic potential energy. Misconception This question highlighted a possible common misconception that when a spring or object is compressed, there is a decrease in the change of elastic potential energy. This is probably as a result that a compression results in a decrease in the length of an object which candidates then relate to a decrease in the change in elastic potential energy. Even though an object is compressed the change in potential energy will always be positive as work is done by a force in the same direction as the displacement.
			Total	1	

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Question			Answer/Indicative content	Marks	Guidance
12			D	1	Examiner's Comments Candidates performed less well on this question. By equating moments, candidates could determine that $x \propto 1/F$. The most common distractor was answer C where candidates incorrectly equated the weight of W to the moment of force *F*.
			Total	1	
13			D	1	Examiner's Comments Only half of candidates answered this question correctly, D, with the most common distractor being answer B. This response demonstrated that they had calculated the correct work done in lifting the motor and then calculated the useful power rather than the power supplied to the motor.
			Total	1	

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Question			Answer/Indicative content	Marks	Guidance
14			**Level 3 (5-6 marks)** Clear description of method to measure *h* and *t* **and** graph analysed to determine *g* and the percentage uncertainty in *g* *There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.* **Level 2 (3-4 marks)** Some description of method to measure *h* and *t* **and** analysis of graph attempted to determine *g* and percentage uncertainty in *g* **or** Clear description of method to measure *h* and *t* **and** limited analysis of graph to determine *g* **or** Limited description of method to measure *h* or *t* **and** graph analysed to determine *g* and the percentage uncertainty in *g* *There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence.* **Level 1 (1-2 marks)** Limited description of the method to measure *h* or *t* **or** Limited analysis to determine *g* *There is an attempt at a logical structure with a line of reasoning. The information is in the most part relevant.* **0 mark** *No response or no response worthy of credit.*	B1 x 6	Use level of response annotations in RM Assessor **Indicative scientific points may include:** **Description of method to** measure *h* and *t* • Use of <u>metre</u> rule(r) / tape measure (not ruler) • Place rule in retort stand • Use of set square / fiducial marker • Timer (or datalogger / computer with detail) connected to electromagnet / trapdoor • Switch off electromagnet to start timer and drop ball • When ball hits trapdoor timer is stopped. • Allow for diameter of ball in height measurement • Resolution of instruments millimetre /millisecond **Ignore** light gates, video **Analysis of data** • Gradient = $\sqrt{\frac{2}{g}}$ or $g = \frac{2}{\text{gradient}^2}$ • Evidence of method of determining gradient • Gradient in the range 0.44 to 0.47 • Determines *g* (≈9.5 m s^{-2}) • Correct power of ten and unit • Draws worst acceptable line • Determines gradient of worst acceptable line • Calculates absolute uncertainty in gradient • Determines *g* from worst acceptable line • Determines percentage uncertainty in gradient • Percentage uncertainty in *g* either 2 × percentage uncertainty in gradient or from *g* values <u>Examiner's Comments</u> This question was designed to test candidates' understanding of practical techniques both designing an experiment and analysing results. High scoring candidates described measuring *h* using a metre rule or tape

Question			Answer/Indicative content	Marks	Guidance
					measure and allowed for the diameter of the ball. Many candidates were unable to explain the use of the electromagnet to release the ball. Some low scoring candidates suggested using a stopwatch. Since the time measurements were recorded to the nearest millisecond it was expected that candidates would describe how the electromagnet and light gate would connect to an electronic timer or datalogger. For the analysis, candidates were expected to link the given equation to the equation of a straight line and thus identify how *g* was related to the gradient. The next logical step would then be to calculate the gradient. For this, it was expected that candidates would demonstrate substituting values from the line on the graph (not data points from the table) to determine the gradient and thus calculate a value of *g* with an appropriate unit. To determine percentage uncertainty, candidate needed to draw the worst acceptable line. This should be either the steepest or shallowest line that passes within all the error bars. Candidates then needed to calculate the worst acceptable gradient. Candidates gained credit for either calculating the percentage uncertainty in *g* from twice the percentage uncertainty in the gradient or from calculating worst value of *g* and then determining the percentage uncertainty. Assessment for learning Candidates should have the opportunity to practise determining values for constants using the gradient and *y*-intercept of straight-line graphs. Candidates should have the opportunity to practise drawing worst acceptable straight lines through error bars and understand the techniques to determine uncertainties in calculated constants using the worst acceptable gradient and/or *y*-intercept.
			Total	6	

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Question			Answer/Indicative content	Marks	Guidance
15	a		Moment = 5 × 9.81 × 0.6 (=29.4) 29.4 N m	C1 A1	**Allow** 29 2sf/30 1sf <u>Examiner's Comments</u> Candidates performed well on this question as they correctly applied the equation moment = force x perpendicular distance from the pivot to calculate the moment as 29.4 Nm and most also gave the correct unit for moment as Nm.
	b		perpendicular (or horizontal) distance between the line of action of the weight and the hips is reduced AW the moment is reduced / very small (so) the force (on the spine) is reduced	B1 M1 A1	**Allow** fulcrum/pivot/turning point for hips **Allow** Keeps the load close to the hips/point H/ reduces (perpendicular) distance to the hips/point H **Allow** turning effect **Ignore** ref. to position of the centre of mass changing **Ignore** ref. to the knees/elbows as the pivot <u>Examiner's Comments</u> There were not many successful responses for this question and many were given 0 marks. Most responses were often vague and unspecific descriptions of a change in the centre of mass, or a distribution of the weight of the box. Most candidates did not establish that by bending the knees, the perpendicular distance to the pivot/hips decreased. Hence the moment about the pivot/hips would be reduced, which in turn reduced the force acting on the spine, resulting in less damage. Some candidates did describe that the moment about the pivot/hips would be reduced by bending the knees. However, a significant number of candidates would then follow this by explaining that there would be 'less stress' on the spine, rather than less force acting on the spine.
			Total	5	

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Question			Answer/Indicative content	Marks	Guidance
16		i	$E = \frac{\sigma}{\varepsilon}$ and F = kx and $\sigma = \frac{F}{A}$ and $\varepsilon = \frac{x}{L}$ Clear substitution leading to $E = \frac{kL}{A}$	M1 A1	ALLOW E = F/A ÷ x/l and F =kx <u>Examiner's Comments</u> Over half of candidates correctly selected the equations *E = σ/ε, ε = x/L, σ = F/A and F = kx* and showed clear substitution to give the expression for the Young modulus *E.* Some candidates did confuse the quantity of *E* and selected the incorrect equation $E = ½ Fx$; $E = ½kx^2$ from the Data, Formulae and Relationships booklet.
		ii	$E = \frac{1670 \times 2.0}{2.9 \times 10^{-8}}$ 1.2×10^{11} N m^{-2} (Uncertainty = 0.05 + 2 + 1.25) 3.3%	C1 A1 B1	ALLOW 3sf 1.15×10^{11} N m^{-2} <u>Examiner's Comments</u> Candidates performed well on this question with about 90% of candidates correctly calculating a value for the Young modulus. About two thirds of candidates correctly determined the percentage uncertainty by calculating the sum of the individual percentage uncertainty values for each of the results.

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Question			Answer/Indicative content	Marks	Guidance
		iii	% difference = $\frac{(1.17-1.20)\times10^{11}}{1.17\times10^{11}}$ = 2.6% % diff < % uncertainty so consistent / accurate / AW	M1 A1	**ALLOW** other quantitative comparisons e.g. 1.11 to 1.19 using 1.15, and 1.16 to 1.24 using 1.2 all within range of 1.17. **ALLOW** calculation using 3 s.f. answer from (ii) to give 1.7% **ALLOW ECF** for correct calculation using uncertainty value from 21c(ii) <u>Examiner's Comments</u> Candidates performed less well on this question, over half the candidates did not achieve marks. Candidates were required to make a quantitative comparison to determine that the researched value was consistent with their calculated value from 21(c)(ii). A significant number of candidates gave a qualitative comparison or simply calculated the difference between the two values to state that the researched value was consistent with their answer to (c)(ii). **Misconception** When attempting to calculate a percentage difference to compare the calculated and researched 'true' value candidates would often find the difference in the calculated and researched value i.e. 0.02×10^{11} and divide by the calculated value of 1.15×10^{11}. Candidates would obtain a decimal value, but it would not be a correct percentage difference. The method to calculate a percentage difference is $\frac{(\text{measured value} - \text{accepted value})}{\text{accepted value}} \times 100\%$ Refer to page 47 in the Practical Skills Handbook on correct methodology on calculating percentage difference for uncertainty in measured and accepted values.
			Total	7	

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Question			Answer/Indicative content	Marks	Guidance
17		i	callipers	B1	<u>Examiner's Comments</u> Many candidates stated a ruler or metre rule without realising that these measuring instruments had a resolution of 1 mm not 0.1 mm. Other candidates incorrectly stated micrometer – a micrometer is not usually used for distances greater than two or three centimetres and normally a micrometer has a resolution of 0.01 mm or better. **Assessment for learning** Candidates should experience a wide range of measuring instruments in a science laboratory.
		ii	32.7 ± 0.2 (mm)	B1	<u>Examiner's Comments</u> Most candidates correctly determined *x* but a significant minority of candidates did not realise that the uncertainty in the measurements needed to be added. **Assessment for learning** Candidates should understand that when quantities are added or subtracted the absolute uncertainties are added. **OCR support** In the Practical Skills Handbook there is a section on uncertainties.

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Question			Answer/Indicative content	Marks	Guidance
		iii	18 ×0.0327 $(\frac{18 \times 0.0327}{9.81} =)$ 0.060 (2sf)	C1 A1	**ALLOW ECF** from **(a)(ii)** **ALLOW** 36 ×0.01635 (alternative method; 1 spring) **Note** Answer must be 2 sf **ALLOW** one mark for 0.06 (1sf) **ALLOW** one mark for 60 (power of ten error) <u>Examiner's Comments</u> The common error was to use 36 N m^{-1} as the force constant and 0.0327 m as the extension. Other errors included either rounding the answer to one significant figure (0.06) or not changing the millimetre to metre. Clear working was needed to allow error carried forward marks into the next section. **Misconception** A number of candidates did not understand significant figures. Candidates should understand the implication of trailing zeros both before and after the decimal place. For example, 0.06 is one significant figure, 0.060 is two significant figures, etc.
		iv	$W = \frac{1}{2} \times 18 \times 0.0327^2$ 9.6 ×10^{-3} (J)	C1 A1	**ALLOW ECF** from **(a)(ii)** and **(iii)** for POT and/or *k* E.g. For *x* = 32.7: 9.6 ×10^3 (J) For *k* = 36 Nm^{-1}: 19.2 ×10^{-3} (J) For *x* = 32.7 and *k* = 36 N m^{-1}: 19.2 ×10^3(J) <u>Examiner's Comments</u> Most candidates gained credit in this question. Many correctly used $W = 0.5kx^2$. Other candidates correctly used $W = 0.5Fx$ using their value for *F* from the previous part. Some lower scoring candidates incorrectly determined the change in gravitational potential energy.
			Total	6	

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Question			Answer/Indicative content	Marks	Guidance
18			3.6 × 0.58 / 2.09 (m) target distance = 1.85 m to 2.15 m **or** 2.09 – 2.0 = 0.09 m = 9 cm 1.85 < 2.09 < 2.15 / 2.00 < 2.09 < 2.15 **or** 9 cm < 15 cm	M1 C1 A1	This mark can be implied by the A1 mark. <u>Examiner's Comments</u> Most candidates correctly calculated that the ball landed at a distance of 2.088 m from the wall and then compared this to the range of the target ring from the wall as 2.0 m ± 0.15 m to show that the child won the ring. Where candidates did not score full marks they had not clearly compared where the ball had landed with the full range of the target ring.
			Total	3	

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